SIMATS ENGINEERING

Assessment Tool 1 (AT1) - Analytical Problem Solving

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Course Code	CSA0709
Course Name	Computer Networks
Course Outcome	CO2 - Configure IP addressing schemes, subnetting and basic routing techniques using a simulation tool, for efficient network design and topology (BL3)
Total	5 questions x 8 marks = 40 marks

Question 1 (8 marks) - Subnetting a Class C network (FLSM)

An organisation is allotted the network **192.168.10.0/24**. It needs **6 equal-sized subnets** for six labs. Determine the new subnet mask, number of subnets and hosts per subnet, and tabulate the network address, usable host range and broadcast address of each required subnet.

Answer

Step 1 - Borrow bits (2 marks). Subnets needed = 6, so 2^n >= 6 gives n = 3 (2^3 = 8 subnets, 2 spare). The mask goes from /24 to **/27**.

Step 2 - Mask and hosts (2 marks). /27 = 255.255.255.224. Host bits = 5, so hosts per subnet = 2^5 - 2 = **30**. Block size = 256 - 224 = **32**.

Step 3 - Subnet table (3 marks).

Subnet	Network	First host	Last host	Broadcast
1	192.168.10.0	192.168.10.1	192.168.10.30	192.168.10.31
2	192.168.10.32	192.168.10.33	192.168.10.62	192.168.10.63
3	192.168.10.64	192.168.10.65	192.168.10.94	192.168.10.95
4	192.168.10.96	192.168.10.97	192.168.10.126	192.168.10.127
5	192.168.10.128	192.168.10.129	192.168.10.158	192.168.10.159
6	192.168.10.160	192.168.10.161	192.168.10.190	192.168.10.191

Step 4 - Analysis (1 mark). Subnets 7 (192.168.10.192/27) and 8 (192.168.10.224/27) remain free for future growth. Total usable hosts = 6 x 30 = 180, versus 254 in the undivided network; the loss (network and broadcast addresses per subnet) is the price of segmentation, which gives smaller broadcast domains and better security.

Question 2 (8 marks) - Analysing a host address in a Class B network (CIDR)

A host has the address **172.16.45.200/20**. Find (a) the subnet mask, (b) the network address, (c) the broadcast address, (d) the usable host range, (e) the number of usable hosts, and (f) whether the host 172.16.50.10 belongs to the same subnet.

Answer

(a) Mask (1 mark). /20 = 11111111.11111111.**1111**0000.00000000 = **255.255.240.0**.

(b) Network address (2 marks). Third octet: 45 = 00101101; mask 240 = 11110000. AND gives 00100000 = 32. Network = **172.16.32.0/20**.

(c) Broadcast (2 marks). Set all 12 host bits to 1: third octet = 0010**1111** = 47, fourth = 255. Broadcast = **172.16.47.255**.

(d) Host range (1 mark). 172.16.32.1 to 172.16.47.254.

(e) Usable hosts (1 mark). $2^{12} - 2 =$ **4094**.

(f) Same subnet? (1 mark). 172.16.50.10 has third octet 50, which is greater than 47, so it lies in the next block (172.16.48.0/20). It is **not** in the same subnet, and a router is required for the two hosts to communicate.

Question 3 (8 marks) - Variable Length Subnet Masking (VLSM) design

A company has the block **192.168.1.0/24** and the following requirements: Dept A - 100 hosts, Dept B - 50 hosts, Dept C - 25 hosts, Dept D - 10 hosts, and one router-to-router serial link (2 hosts). Design an efficient addressing plan using VLSM and show that fixed-length subnetting cannot satisfy it.

Answer

Method (2 marks). Sort requirements from largest to smallest and allocate each from the next free address, using the smallest block that fits (2^h - 2 >= hosts).

Allocation (4 marks).

Segment	Hosts	Block / Mask	Network	Usable range	Broadcast
Dept A	100	/25 (128) 255.255.255.128	192.168.1.0	.1 - .126	192.168.1.127
Dept B	50	/26 (64) 255.255.255.192	192.168.1.128	.129 - .190	192.168.1.191
Dept C	25	/27 (32) 255.255.255.224	192.168.1.192	.193 - .222	192.168.1.223
Dept D	10	/28 (16) 255.255.255.240	192.168.1.224	.225 - .238	192.168.1.239
WAN link	2	/30 (4) 255.255.255.252	192.168.1.240	.241 - .242	192.168.1.243

Analysis (2 marks). Addresses used = 128 + 64 + 32 + 16 + 4 = 244 of 256, leaving 192.168.1.244 - .255 (12 addresses) free. With FLSM, the largest department (100 hosts) forces every subnet to /25, which yields only 2 subnets in a /24, so five segments cannot be fitted. VLSM therefore saves address space and supports the design.

Question 4 (8 marks) - Static routing configuration and verification

Three routers are connected in a line: R1 - R2 - R3. LANs: R1 = 192.168.1.0/24, R2 = 192.168.2.0/24, R3 = 192.168.3.0/24. Links: R1-R2 = 10.0.0.0/30 (R1 .1, R2 .2); R2-R3 = 10.0.0.4/30 (R2 .5, R3 .6). Write the static routes needed for full connectivity, suggest an optimisation, and state how you would verify it in a simulator such as Cisco Packet Tracer.

Answer

Analysis (2 marks). Each router knows only its directly connected networks. R1 must learn 192.168.2.0, 192.168.3.0 and 10.0.0.4/30; R2 must learn 192.168.1.0 and 192.168.3.0; R3 must learn 192.168.1.0, 192.168.2.0 and 10.0.0.0/30. Next hop is always the neighbour router on the shared link.

Configuration (3 marks).

```
R1(config)# ip route 192.168.2.0 255.255.255.0 10.0.0.2
R1(config)# ip route 192.168.3.0 255.255.255.0 10.0.0.2
R1(config)# ip route 10.0.0.4   255.255.255.252 10.0.0.2

R2(config)# ip route 192.168.1.0 255.255.255.0 10.0.0.1
R2(config)# ip route 192.168.3.0 255.255.255.0 10.0.0.6

R3(config)# ip route 192.168.1.0 255.255.255.0 10.0.0.5
R3(config)# ip route 192.168.2.0 255.255.255.0 10.0.0.5
R3(config)# ip route 10.0.0.0   255.255.255.252 10.0.0.5
```

Optimisation (1.5 marks). R1 and R3 are stub routers, so each can use a single default route instead: `ip route 0.0.0.0 0.0.0.0 10.0.0.2` on R1 and `ip route 0.0.0.0 0.0.0.0 10.0.0.5` on R3. This reduces the routing table to one entry per stub router.

Verification (1.5 marks). Use `show ip route` (routes appear with code 'S'), then `ping 192.168.3.10` from a PC in 192.168.1.0 and `tracert` to confirm the path R1 - R2 - R3. Limitation: static routes do not adapt to link failure and become hard to maintain as the network grows, which motivates dynamic routing (Question 5).

Question 5 (8 marks) - Dynamic routing: RIP v2 vs OSPF

For the same topology as Question 4, the R1-R2 link is FastEthernet (100 Mbps), the R2-R3 link is a serial link of 1.544 Mbps, and the R3 LAN is FastEthernet. (a) Configure RIPv2 and OSPF on R2. (b) Compute the RIP metric and the OSPF cost from R1 to 192.168.3.0/24. (c) Compare the two protocols and justify a choice.

Answer

(a) Configuration on R2 (3 marks).

```
! RIP version 2
R2(config)# router rip
R2(config-router)# version 2
R2(config-router)# no auto-summary
R2(config-router)# network 192.168.2.0
R2(config-router)# network 10.0.0.0

! OSPF (single area 0)
R2(config)# router ospf 1
R2(config-router)# network 192.168.2.0 0.0.0.255 area 0
R2(config-router)# network 10.0.0.0   0.0.0.3   area 0
R2(config-router)# network 10.0.0.4   0.0.0.3   area 0
```

(b) Metrics (2.5 marks). RIP uses hop count: R1 to R2 to R3 gives metric **2** to 192.168.3.0 (maximum usable is 15; 16 means unreachable). **OSPF** cost = reference bandwidth / interface bandwidth, with the default reference of 100 Mbps. FastEthernet = 100/100 = 1; serial 1.544 Mbps = 100,000,000 / 1,544,000 = 64.77, truncated to 64. Cost from R1 = 1 (R1-R2) + 64 (R2-R3) + 1 (R3 LAN) = **66**.

(c) Comparison (2.5 marks).

Feature	RIPv2	OSPF
Type	Distance vector	Link state
Metric	Hop count (ignores bandwidth)	Cost based on bandwidth
Convergence	Slow (periodic updates every 30 s)	Fast (event-triggered LSAs)
Scalability	Limited to 15 hops	Large networks, area hierarchy
Administrative distance	120	110
Overhead / complexity	Low, simple to configure	Higher CPU/memory, more complex

Justification. RIP would treat the slow serial link and a fast link as equal (1 hop), so it can choose a poorer path. OSPF accounts for bandwidth and converges faster, so it is the better choice for anything beyond a small network, while RIPv2 suffices for a small, simple lab topology.

Marks summary: Q1 8 + Q2 8 + Q3 8 + Q4 8 + Q5 8 = 40 marks.